

A. Project Overview

- **Goal:** Visualizing and mapping passing data for a soccer dataset. Includes key functions for important information. Function outputs gets printed in terminal. Visualized maps are saved in outputs folder.
- **Dataset:** <https://github.com/statsbomb>

B. Data Processing

- From that github link above, I used python to clean up the data set to pull out the specific game that I wanted to analyse. I then exported data as 2 csvs, one for passing data and one for player data. I then read the files using methods taught in class in Rust

C. Code Structure

1. Modules

- key_functions.rs: consists of all key functions
- main.rs: main interactive interface of program
- mapping.rs: visualizing/ mapping data, saves everything to output folder
- players.rs: reads the players data
- raw_passes: reads the passes data

2. Key Functions & Types (Structs, Enums, Traits, etc)

- **fn top_passes**
 - **function:** prints all player's passing partners in order of most to least.
 - **Inputs:** player name, passing data, and a bool on whether or not to print
 - **logic:** iterates through data and only pushes data into HashMap if name is correct and recipient name isn't empty. The loop with if print determines if it prints. HashMap has to be iterated into a vector so that it can be ordered
- **fn starting_xi**
 - **function:** prints and returns starting eleven players
 - **inputs:** team name and player data
 - **logic:** makes sure team name matches and player is a starter, then pushes it into hashmap and prints out the name and jersey number
- **fn average_pass_start_location**
 - **function:** averages out a player's passing start location
 - **inputs:** passer name, passing data
 - **logic:** set a mutable x and y total value and pass count. Every time passer name matches and pass is successful the values are added. At the end I do total/pass count to find averages
- **fn pass_success**
 - **function:** calculates a player's pass success %
 - **inputs:** passer name and passing data
 - **logic:** has a mutable counter for success and total updates total when pass is unsuccessful and both when it is, the last if loop is to make sure that a valid name is entered, if its invalid player name is prints out error
- **fn most_passes**
 - **function:** prints the team's top passers, returns a HashMap of player and passes count
 - **inputs:** team name and the passing data
 - **logic:** iterates through every line but only pushes data into HashMap if the name is correct and outcome name is empty, uses iterator to iterate into vector so it can be ordered and printed
- **fn draw_field**
 - **function:** draws out the field
 - **inputs:** filename
 - **logic:** uses plotter to draw every line
- **fn player_passing_map**
 - **functions:** returns a png that maps out a player's individual passing map, with red arrows indicating failed passes and green indicating successful

- **inputs:** player name and passing data
- **logic:** Calls draw field function to initialize a field. Need to create x_scale and y_scale because given data isn't in 800x600 dimensions it's in 120x100. I iterate over data only letting data through if name matches. Draw lines and arrow head (using trig) for passes. Green= successful pass, red= failed pass
 - **fn team_passing**
- **function:** returns a png that maps out average team passing map, linking players together if they have more than 5 passes and increasing arrow size as # of passes grow
- **inputs:** team name and passing data
- **logic:** I call draw_field and apply the same x and y scaling. I initialize the team's starters using starting_xi since only they will be mapped. For each starter, I call average_pass_start_location to get their average position and top_passes to find their main recipients. I check that recipients are also starters to exclude subs. I scale the arrow width based on pass count. The arrow endpoint is shifted slightly using trigonometry to prevent overlap with player bubbles. Finally, I draw the starter bubbles and label them with their jersey numbers.
 - **struct PlayerData**
- stores each row of player data: contains player id, name, team, starter or not, jersey number
- **struct PassingData**
- stores each row of passing data: contains minute, player_id, player_name, team_name, x, y, end_x, end_y, pass_recipient_id, pass_recipient_name, outcome_id, outcome_name

3. Main Workflow

- o All functions are called through interactive main interface

D. Tests

```
running 5 tests
test key_functions::tests::test_most_passes ... ok
test key_functions::tests::test_average_pass_start_location_basic ... ok
test key_functions::tests::test_top_passes_basic ... ok
test key_functions::tests::test_starting_xi ... ok
test key_functions::tests::test_pass_success ... ok

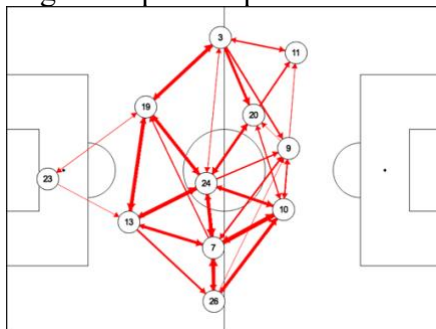
test result: ok. 5 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
```

Each test tests a function under key_function, instead of a standardized data set for all tests, each test has its own data so it's easier to understand as only the required data will be in the set data set.

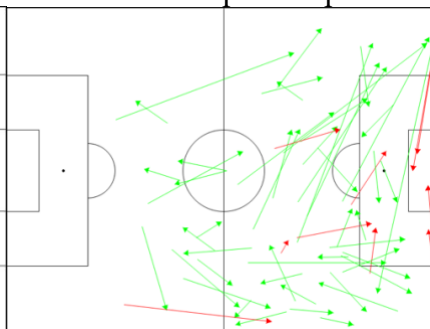
E. Results

- All program outputs: There are many, here are some examples:

Argentina pass map



Messi's own pass map



```
Welcome to France vs Argentina 2022 WC Final Stats
Choose an Option (Type Option Number):
1. Team Starting Lineups
2. Player's Top Passing Partners
3. Player Passing Map
4. Team Passing Map
5. Player Passing Percentage
6. Team's Top Passers
7. Exit
```

Top Passers on Argentina	Count
Passer	79
Enzo Fernandez	79
Nicolás Hernán Otamendi	69
Cristian Gabriel Romero	60
Rodrigo Javier De Paul	57
Lionel Andrés Messi Cuccittini	49
Alexis Mac Allister	45
Nicolás Alejandro Tagliafico	44
Nahuel Molina Lucero	39
Julían Álvarez	26
Ángel Fabián Di María Hernández	25
Marcos Javier Acuña	19
Damián Emiliano Martínez	18
Leandro Daniel Paredes	15
Gonzalo Ariel Montiel	8
Lautaro Javier Martínez	5
Germán Alejandro Pezzella	1
Paulo Bruno Exequiel Dybala	1

Argentina Starting XI		Passer	Recipient	Count
Name	Jersey Number			
Nahuel Molina Lucero	26	Lionel Andrés Messi Cuccittini	Rodrigo Javier De Paul	13
Rodrigo Javier De Paul	7	Lionel Andrés Messi Cuccittini	Enzo Fernandez	8
Cristian Gabriel Romero	13	Lionel Andrés Messi Cuccittini	Julían Álvarez	8
Nicolás Hernán Otamendi	19	Lionel Andrés Messi Cuccittini	Alexis Mac Allister	6
Nicolás Alejandro Tagliafico	3	Lionel Andrés Messi Cuccittini	Ángel Fabián Di María Hernández	5
Alexis Mac Allister	20	Lionel Andrés Messi Cuccittini	Nahuel Molina Lucero	5
Damián Emiliano Martínez	23	Lionel Andrés Messi Cuccittini	Marcos Javier Acuña	2
Lionel Andrés Messi Cuccittini	10	Lionel Andrés Messi Cuccittini	Cristian Gabriel Romero	2
Ángel Fabián Di María Hernández	11	Lionel Andrés Messi Cuccittini	Lautaro Javier Martínez	2
Julían Álvarez	9	Lionel Andrés Messi Cuccittini	Gonzalo Ariel Montiel	2
Enzo Fernandez	24	Lionel Andrés Messi Cuccittini	Nicolás Hernán Otamendi	1

Lionel Andrés Messi Cuccittini's pass percentage: 81%

F. Usage Instructions

- **How to build and run your code:** Cargo run
- **Description of any command-line arguments or user interaction in the terminal:**

After cargo run, user will be prompted with interface with 7 options. Type in a number for your choice. Follow input prompt. If asked for name, full name is required, it is recommended to call starting lineup first for everyone's full name then copy name for other function calls. Ex: cannot just type Messi need to type Lionel Andrés Messi Cuccittini. For the passing maps, results will be returned into output folder, interface should tell you what the file name is called.

When you type cargo run:

```
Welcome to France vs Argentina 2022 WC Final Stats
Choose an Option (Type Option Number):
1. Team Starting Lineups
2. Player's Top Passing Partners
3. Player Passing Map
4. Team Passing Map
5. Player Passing Percentage
6. Team's Top Passers
7. Exit
```

EX: 1 then enter team name and hit enter

```
1
Enter team:
Argentina
```

Returns:

```
Argentina Starting XI
Name                               Jersey Number
Nahuel Molina Lucero              26
Rodrigo Javier De Paul             7
Cristian Gabriel Romero            13
Nicolás Hernán Otamendi            19
Nicolás Alejandro Tagliafico       3
Alexis Mac Allister                20
Damián Emiliano Martínez            23
Lionel Andrés Messi Cuccittini      10
Ángel Fabián Di María Hernández    11
Julían Álvarez                     9
Enzo Fernandez                     24
```